

Multi-Model Genre Classification and Book-to-Movie Mismatch Detection

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1 Introduction

Genre classification is a fundamental task in natural language processing (NLP) that enables automated content organization, recommendation systems, and literary analysis. Unlike single-label classification, real-world texts frequently belong to multiple genres simultaneously; a novel may be classified as both *Thriller* and *Sci-Fi*, making multi-label classification the appropriate formulation. However, this setting introduces significant challenges, including label co-occurrence dependencies, class imbalance across genres, and subjective genre boundaries that vary across annotators and corpora.

In this work, we address two interconnected tasks. First, we develop a multi-label genre classification system covering 12 canonical genres by fine-tuning four transformer-based architectures: DistilBERT, DeBERTa-v3-base, RoBERTa-base, and Longformer. We further investigate whether incorporating latent topic information through a fusion architecture can improve classification performance. Second, we build a downstream application that leverages the trained classifier to detect narrative mismatches between books and their movie adaptations, using a novel five-head composite scoring system applied to Wikipedia-sourced plot data.

Our primary contributions are as follows:

- A systematic comparison of four transformer architectures on multi-label genre classification with per-genre threshold optimization.
- A topic-aware fusion model that concatenates RBF-kernel topic probabilities with transformer representations.
- An end-to-end book-to-movie mismatch detection pipeline comprising Wikipedia scraping, multi-head scoring, and automated quality filtering.

2 Dataset

2.1 Data Sources

We combine two publicly available datasets for the genre classification task. The CMU Book Summaries Corpus¹ provides 16,559 book summaries with Freebase genre annotations. The Kaggle Movies Dataset² contributes 45,466 movie metadata records including plot overviews and TMDb genre tags. By merging these two sources, we obtain a diverse corpus that spans both literary and cinematic texts.

2.2 Preprocessing

The raw genre labels from both sources exhibit considerable fragmentation, with over 300 unique genre strings across the two corpora. We consolidate these into a unified taxonomy of 12 canonical genres: *Action*, *Comedy*, *Drama*, *Horror*, *Thriller*, *Sci-Fi*, *Fantasy*, *Romance*, *Animation*, *Documentary*, *Literary Fiction*, and *Non-Fiction*. This mapping is performed through a combination of direct string matching and manual inspection of ambiguous labels. Records with no valid genre mapping after consolidation

¹<https://www.cs.cmu.edu/~dbamman/booksummaries.html>

²<https://www.kaggle.com/datasets/rounakbanik/the-movies-dataset>

are discarded. Genre labels are then binarized using a `MultiLabelBinarizer`, producing a binary label matrix suitable for multi-label training.

2.3 Data Splits and Statistics

We apply iterative stratified splitting [8] to partition the data into training, validation, and test sets with a 70/15/15 ratio. Iterative stratification preserves the label distribution across all 12 genres in each split, which is critical given the significant class imbalance present in the dataset.

Table 1: Dataset split statistics.

Split	Samples	Percentage
Training	36,104	70%
Validation	7,774	15%
Test	7,740	15%
Total	51,618	100%

The most frequent genre is *Drama* (3,135 test samples), while the least frequent is *Non-Fiction* (64 test samples), yielding an imbalance ratio of approximately 49:1. This imbalance motivates our use of positive-weight-adjusted loss functions and per-genre threshold tuning.

3 Approach

3.1 Model Selection

We select four transformer architectures that represent different trade-offs between computational efficiency, contextual depth, and input length capacity.

DistilBERT [1] is a distilled variant of BERT that retains 97% of BERT’s language understanding while being 60% faster and 40% smaller. It serves as our efficiency baseline.

DeBERTa-v3-base [2] introduces disentangled attention, which separately encodes content and position information using two independent vectors. This architecture has demonstrated state-of-the-art results on several NLU benchmarks.

RoBERTa-base [3] improves upon BERT through optimized pretraining, including dynamic masking, larger mini-batches, and removal of the next-sentence prediction objective. Its robust pretraining makes it well-suited for downstream fine-tuning tasks.

Longformer [4] extends the standard transformer with a combination of local sliding-window attention and global attention, enabling efficient processing of documents up to 4,096 tokens. This is particularly relevant for our corpus, where book summaries can exceed the 512-token limit of standard models.

3.2 Topic Modeling Fusion

To investigate whether latent thematic information can complement transformer representations, we design a two-stream fusion architecture. We first train a topic model using `MiniBatchKMeans` with $k = 30$ clusters on TF-IDF representations, applying a Gaussian RBF kernel with calibrated bandwidth σ to obtain soft topic probability vectors for each document.

The fusion classifier processes two parallel streams:

1. The `[CLS]` token representation from RoBERTa (768 dimensions) is projected through a linear layer to 256 dimensions, followed by LayerNorm, GELU activation, and dropout (0.3).
2. The 30-dimensional topic probability vector is projected to 64 dimensions with its own LayerNorm, GELU, and dropout (0.2).

The two streams are concatenated into a 320-dimensional vector and passed through a final classification head. Independent LayerNorm layers ensure scale alignment between the two representation spaces.

3.3 Training Strategy

All models are trained with binary cross-entropy loss with logits (`BCEWithLogitsLoss`), incorporating class-specific positive weights to address label imbalance. We use the AdamW optimizer with a cosine learning rate scheduler and 10% linear warmup.

A key component of our approach is **per-genre threshold tuning**. Rather than applying a fixed threshold of 0.5 to the sigmoid outputs, we perform an independent sweep over the range $[0.20, 0.85]$ for each genre, selecting the threshold that maximizes the per-genre F1 score on the validation set. This approach consistently improves Macro-F1 by 3–5 points across all models compared to a static threshold.

For memory efficiency, all datasets are pre-tokenized and cached directly in GPU VRAM using a custom `UltraFastGenreDataset` class. Gradient checkpointing is enabled for RoBERTa and DeBERTa to allow batch sizes of 64 within the 15.8 GB VRAM budget of the Tesla T4 GPU. Mixed-precision training (FP16) is used for all models except DeBERTa-v3, which requires FP32 due to numerical instability in its disentangled attention computation under reduced precision.

For the fusion model, we freeze the RoBERTa backbone during the first epoch to prevent pretrained gradients from overwhelming the randomly initialized topic head. After the first epoch, we unfreeze the backbone and apply differential learning rates: 1×10^{-5} for the RoBERTa body and 5×10^{-5} for the fusion head.

4 Implementation Details

4.1 Training Environment

All transformer models are trained on Kaggle and Google Colab instances equipped with NVIDIA Tesla T4 GPUs (15.8 GB VRAM). The software environment consists of Python 3.12, PyTorch 2.10, and HuggingFace Transformers 4.48+. Data preprocessing and topic model training are performed locally.

4.2 Classification Pipeline

The classification pipeline proceeds as follows. Raw text is tokenized using each model’s respective tokenizer with a maximum sequence length of 512 tokens (256 for DeBERTa due to VRAM constraints). The tokenized inputs, attention masks, and binary label vectors are packed into tensors and transferred to GPU VRAM before training begins. During training, we perform forward passes with optional gradient checkpointing, compute the weighted BCE loss, and update parameters using AdamW with gradient scaling for FP16 models. At the end of each epoch, we evaluate on the validation set with per-genre threshold tuning and save the model checkpoint if the Macro-F1 score improves. Early stopping with a patience of 5 epochs prevents overfitting.

4.3 Wikipedia Scraping Pipeline

For the mismatch detection task, we develop a multi-pass Wikipedia scraper to identify book-to-movie adaptation pairs and retrieve their respective plot texts. The scraper operates in four passes with decreasing confidence:

1. **Gold (Wikidata P144):** Batch SPARQL queries retrieve films linked to books via the “based on” (P144) property. This yields the highest-confidence matches.
2. **Silver (Book page):** The scraper parses the “Adaptations” sections of book Wikipedia articles to find linked film pages.
3. **Silver (Author category):** Author category pages are crawled for film adaptation references.

4. **Bronze (Fuzzy matching):** Title-based fuzzy matching with verification through content overlap analysis.

This pipeline produces 619 book-movie pairs with full plot texts. A confidence score is assigned to each pair based on its verification tier.

4.4 Mismatch Detection

We design a five-head composite scoring system that quantifies narrative divergence between book summaries and movie plots from multiple perspectives:

Table 2: Mismatch detection scoring heads and weights.

Head	Metric	Weight
Semantic Distance	Cosine distance (MiniLM-L6-v2 embeddings)	0.35
Genre Shift	Jensen-Shannon Divergence of RoBERTa genre predictions	0.25
TF-IDF Distance	Cosine distance of TF-IDF vectors (bigram, 15K features)	0.25
Entity Overlap	Jaccard distance of spaCy NER entity sets	0.10
Sentiment Delta	Absolute VADER compound score difference (first 1500 chars)	0.05

The entity overlap weight is reduced to 0.10 due to its high correlation (0.81) with TF-IDF distance, which would otherwise double-count vocabulary-level similarity. Sentiment delta receives a low weight of 0.05 because VADER compound scores exhibit a systematic baseline difference between prose registers (literary vs. encyclopedic), inflating the raw delta.

Before scoring, we apply a **wrong-match filter** that removes pairs with semantic cosine similarity below 0.20. This step eliminates 20 scraper errors (e.g., mismatched titles) that would otherwise pollute the mismatch rankings as false positives.

5 Results and Discussion

5.1 Model Comparison

Table 3 presents the performance of all five model configurations on the held-out test set. RoBERTa-base achieves the highest Macro-F1 of 0.6870, followed closely by Longformer at 0.6882. DeBERTa-v3 underperforms relative to its architectural sophistication, which we attribute to the FP32 constraint limiting effective batch size and training throughput.

Table 3: Test set performance comparison across all models.

Model	Macro-F1	Micro-F1	Hamming Loss	Train Time
DistilBERT	0.6831	0.6875	0.0984	1h 13m
DeBERTa-v3-base	0.6496	0.6689	0.1077	6h 33m
RoBERTa-base	0.6870	0.6951	0.0975	3h 18m
Longformer	0.6882	0.6979	0.0953	6h 46m
RoBERTa + Topic Fusion	0.6828	0.6897	0.1003	2h 47m

5.2 Per-Genre Analysis

Table 4 reports the per-genre precision, recall, and F1 scores for RoBERTa-base. *Documentary* achieves the highest F1 (0.84), likely due to its distinctive vocabulary that is well-separated from fictional genres. *Romance* exhibits the lowest F1 (0.57), reflecting the genre’s frequent co-occurrence with *Drama* and *Comedy*, which introduces label confusion. *Non-Fiction* achieves a strong F1 of 0.70 despite having

only 64 test samples, suggesting that the positive-weight adjustment effectively compensates for extreme class imbalance.

Table 4: Per-genre test results for RoBERTa-base with per-genre threshold tuning.

Genre	Precision	Recall	F1	Support
Action	0.58	0.72	0.64	1,247
Comedy	0.64	0.74	0.69	1,958
Drama	0.67	0.86	0.75	3,135
Horror	0.67	0.76	0.71	785
Thriller	0.67	0.77	0.71	1,913
Sci-Fi	0.77	0.74	0.75	902
Fantasy	0.54	0.70	0.61	710
Romance	0.50	0.66	0.57	1,066
Animation	0.58	0.67	0.62	286
Documentary	0.84	0.85	0.84	581
Literary Fiction	0.51	0.85	0.64	718
Non-Fiction	0.64	0.77	0.70	64
Macro Avg	0.63	0.76	0.69	13,365

5.3 Training Dynamics

Table 5 shows the epoch-by-epoch training progression for RoBERTa-base. The model achieves its best validation Macro-F1 of 0.6951 at epoch 7, after which performance plateaus. Training loss decreases monotonically from 1.2174 to 0.3006, indicating stable convergence without signs of catastrophic overfitting.

Table 5: RoBERTa-base training progression (per-genre threshold tuning applied at each epoch).

Epoch	Training Loss	Val Macro-F1	Model Saved
1	1.2174	0.6533	Yes
2	0.5916	0.6631	Yes
3	0.4891	0.6747	Yes
4	0.4197	0.6824	Yes
5	0.3614	0.6785	No
6	0.3398	0.6888	Yes
7	0.3174	0.6951	Yes
8	0.3035	0.6945	No

5.4 Topic Fusion Analysis

The RoBERTa + Topic Fusion model achieves a test Macro-F1 of 0.6828, which is 0.0042 points below the vanilla RoBERTa baseline. Analysis of per-genre performance reveals that the fusion model improves only on *Fantasy* (+0.035 F1) and *Non-Fiction* (+0.005 F1), while all other genres show marginal regression. This suggests that the [CLS] representation from RoBERTa already captures sufficient thematic information, and the 30-dimensional topic vector introduces noise rather than complementary signal for most genres.

5.5 Inference Examples

To demonstrate the practical utility of the trained classifier, we present three inference examples using the best RoBERTa-base model with per-genre threshold tuning applied.

Table 6: Inference examples from the RoBERTa-base classifier.

Input Text	Predicted Genres (Confidence)
“A cybernetic soldier fights rogue AI controlling a dystopian megacity...”	Sci-Fi (99.85%), Action (96.53%), Animation (63.48%), Thriller (51.66%)
“Two childhood friends reunite years later and rediscover their love...”	Romance (98.29%), Drama (90.33%), Comedy (65.23%)
“An abandoned hospital hides a terrifying supernatural entity...”	Horror (99.76%), Thriller (58.06%), Fantasy (33.54%), Sci-Fi (20.70%)

The model correctly identifies the dominant genres with high confidence and assigns plausible secondary labels. For example, the first input receives strong *Sci-Fi* and *Action* predictions, with a moderate *Thriller* score reflecting the dystopian conflict theme. The third input demonstrates appropriate co-activation of *Horror* and *Fantasy* for supernatural content.

5.6 Mismatch Detection Results

After applying the wrong-match filter, 599 pairs remain for mismatch analysis. The composite mismatch scores follow an approximately normal distribution with mean 0.5015 and standard deviation 0.1602. We partition the scores into three categories using the 33rd and 67th percentiles: *Faithful* (< 0.418 , 198 pairs), *Moderate* (0.418–0.564, 203 pairs), and *Significant* (> 0.564 , 198 pairs).

To validate the scoring system, we examine known adaptations. Faithful adaptations such as *The Big Sleep* (0.212), *To Kill a Mockingbird* (0.245), and *Pride and Prejudice* (0.256) correctly receive low mismatch scores. Divergent adaptations such as *A Clockwork Orange* (0.422) and *I Am Legend* (0.535) receive appropriately elevated scores.

A notable limitation is observed for *The Shining*, which scores 0.327 (Faithful) despite Kubrick’s film being widely regarded as a significant departure from King’s novel. This occurs because entity overlap and TF-IDF similarity remain high (the film shares all major characters, locations, and surface plot elements), while the thematic inversion is captured only by the semantic distance head. This illustrates a known limitation of surface-level metrics for detecting thematic, as opposed to structural, divergence.

The head correlation analysis confirms that semantic distance and TF-IDF distance are the most informative components, with correlations of 0.85 and 0.85 to the composite score, respectively. Genre shift shows moderate correlation (0.57), while sentiment delta contributes minimally (0.22), consistent with its low weight assignment.

6 Improvements and Future Work

Several directions could yield meaningful improvements to both the classification and mismatch detection components.

Larger pretrained models. Scaling to RoBERTa-large (355M parameters) or DeBERTa-v3-large could improve representation quality, particularly for underperforming genres such as *Romance* and *Fantasy*. The primary bottleneck is GPU memory; however, gradient accumulation and 8-bit quantization techniques could make this feasible on consumer hardware.

Cross-attention fusion. The current topic fusion architecture uses simple concatenation to combine transformer and topic representations. A cross attention mechanism that allows the two streams to

attend to each other could enable more nuanced integration, potentially recovering the performance lost to concatenation noise.

Improved topic features. Replacing MiniBatchKMeans with neural topic models such as NVDM or ProdLDA may yield more discriminative topic distributions. Additionally, training the topic model jointly with the classifier in an end-to-end fashion could align topic representations with the classification objective.

Thematic mismatch detection. The current scoring system relies on surface-level features that cannot capture thematic inversions, as demonstrated by *The Shining*. Incorporating sentence-level discourse analysis or narrative arc modeling could address this limitation. Specifically, encoding the progression of emotional valence across narrative segments, rather than computing a single aggregate sentiment score, would better capture structural divergence.

Wrong-match threshold calibration. Lowering the cosine similarity threshold from 0.20 to 0.15 would retain genuine but highly divergent adaptations such as *Heart of Darkness* to *Apocalypse Now* (cosine similarity 0.181), which are currently filtered as false positives.

References

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AID-729 Assignment 4

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Paper 8. Contextual Word Representations: A Contextual Introduction

a) What Problem does the paper Addresses?

This paper addresses the fundamental challenge in NLP i.e. how can we represent words so that the computers can understand their meaning. Earlier the word representations like Word2Vec and GloVe assigns a single and fixed vector to each word regardless of the context. The word like “bank” which can have different word meanings like financial institution or the side of a river depending on the sentence. So, this paper explores the contextual word representations like ELMo and BERT where the word vectors changes depending on the sentence it has appeared.

b) Why was it necessary to address this problem was there any earlier work trying to address the same technology challenges?

It was necessary to address this problem because for all the tasks like translation, question answering, sentiment analysis requires to represent the word in such a way that how words are related to each other.

Earlier people tried different method to represent words, some are simple integer mappings, One-hot encoding, co-occurrence count vectors, and dense word embeddings like Word2Vec and GloVe. All these traditional methods have the same problem i.e. one fixed vector per word and one fixed vector cannot represent the different meaning of a word.

c) What unique mathematical approach the paper adopts?

Unfortunately, this paper is intentionally kept non-mathematical. It has not used any equations or proofs instead it explained the intuitive reasoning behind each method. It talks about the concept “meaning comes from the context”, how co-occurrence matrices captures word relations, how neural language model learn dense vectors by predicting surrounding words and how self-attention allow every word in a sentence to look every other word to build context based representation

d) Do you think the results are satisfactory, do they represent what the work was set out to be?

This is a survey or a tutorial paper. It showed that there is 16% improvement semantic role labelling and 9% on SQuAD question Answering benchmark. Also, this paper is able to convey why contextual representations like ELMo and BERT is better than traditional method of one vector per word. It is able to convey the significance of these advances without having deep ML expertise.

e) Write Your observation on how the work can be improved?

This paper does not discuss the limitations of contextual embeddings i.e. massive computational cost or any bias they absorb from training data. Could have included this.

Author could have also included some section about when to use static embeddings and when to use contextual embedding, the trade off in speed, memory, computation and complexity. Including these could have make this paper desirable for the practitioners.

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- a) **What problem the paper addresses:** The paper tries to solve the problem of training models like classifier with very small dataset also known as few shot learning. It tries to combine the zero-shot prompting in which we feed the model a task description in natural language like "It was ____ . The food was amazing!" (known as PET pattern) and asking the model to predict the blank word and standard supervised training along with large pool of unlabeled examples to bridge the gap between the two approaches with the goal begin to increase the overall accuracy.
- b) **Why was it necessary to address this problem, was there any earlier work trying to address the same technology challenges:** It was necessary to address this problem because in real world there are vast number of different languages and niche domains which might not have labelled data or data itself. Creation of annotated data takes significant amount of resources, standard supervised learning gives poor performance with small training sets, zero shot prompting underperforms the supervised learning are the reason the paper states. The closest work is by Xie et al. which tries to use unlabeled data with small labeled sets and use backtranslation for data augmentation but the translation model itself requires thousands of labeled examples which defeats the goal of few-shot learning.
- c) **What unique mathematical approach the paper adopts:** The most unique mathematical approach used in this paper is converting the classification task into the language modelling score. In classification our goal is to assign a class to an input like positive or negative but in Pattern-Exploiting Training we ask about the probability of a specific word being assigned to the masked position. For a pattern-verbalizer pair (P, v) with input x the score for label 1 is defined as: $s_{p(l) | x} = M(v(l) | P(x))$ where M is the masked language model, $v(l)$ is the word (verbalizer) mapped to label l , $P(x)$ is the cloze-style pattern applied to input x . Therefore instead of classification score we get a language model probability for a specific word. The raw score is then converted into probability distribution by applying softmax: $q_{p(l) | x} = e^{s_{p(l) | x}} / \sum e^{s_{p(l') | x}}$ which gives the valid probability distribution over all labels. Another mathematical unique approach is combined loss function which addresses the catastrophic forgetting, a problem where finetuning on small dataset causes the model to forget its pretrained language knowledge. The paper combines two losses: $L = (1 - \alpha) \cdot L_{CE} + \alpha \cdot L_{MLM}$. This combines classification loss and language modeling loss with a very small $\alpha = 10^{-4}$ to prevent the model from forgetting its pretrained language knowledge while learning from limited labeled examples.
- d) **Do you think the results are satisfactory, do they represent what the work was set out to be:** The core goals of the paper like demonstrating PET outperforming standard supervised training in few-shot setting were satisfactory as evident from table 1 which shows PET and Iterative PET significantly outperforming Supervised training also Table 2 shows the PET and iPET also consistently outperforms UDA and MixText semi-supervised baselines. Another goal was to demonstrate that PET work across multiple task and languages which is also demonstrated by using 4 different datasets covering sentiment, topic classification, QA and inference tasks and the paper observes improvement across all of them. The paper also notes it's limitation like narrowing of performance gap as the dataset size increases, MNLI (evaluating language understanding by classification task) results at small sizes being weak etc.
- e) **Write your observation on how the work can be improved. Feel free to refer to any other subsequent publication that presents a critique to this paper:** The paper relies on humans to write good patterns and verbalizers like "It was ____ ." for sentiment and "Yes/No" for entailment which it tries to solve by using the Automatic Verbalizer Search but experiments show it's performance to be significantly worse than the human written ones. In the paper by Gao et al. "Making Pre-trained Language Models Better Few-shot Learners" tries to solve this problem by automatic prompt generation using the language model itself to search for good patterns. Another observation I made was the single token verbalization bottleneck. The PET's verbalizer maps each label to exactly one token from the vocabulary which is significant constraint because many meaningful label

descriptions need multiple words for example “highly recommend”, and some labels are inherently multi-word concepts which cannot be captured by a single token.